

## ZAKATALA, Azerbaijan

WMO: 375750

Lat: 41.6201N

Lon: 46.6486E

Elev: 1608

StdP: 13.86

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 24.2       | 26.8 | 14.7                          | 12.3 | 31.5 | 17.5 | 14.1 | 32.7 | 11.8                  | 48.3 | 9.7 | 44.1 | 3.9                   | 20   | 0.384 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.8                   | 94.8            | 70.4 | 92.2 | 70.1 | 89.9 | 69.5 | 74.2                | 87.6 | 72.6 | 86.5 | 71.1 | 85.3 | 4.6                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 70.3                            | 118.3 | 80.4 | 68.2 | 110.0 | 80.0 | 66.4 | 103.1 | 79.1 | 38.8          | 87.7 | 37.4 | 86.5 | 36.0 | 85.7 | 82.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |     | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 10.4              | 8.0  | 6.8 | 20.2                       | 100.0 | 4.5                | 2.6 | 17.0                                               | 101.9 | 14.3       | 103.4 | 11.8       | 104.8 | 8.5        | 106.7 |
|                   |      |     | 19.2                       | 76.5  | 4.3                | 2.5 | 16.1                                               | 78.3  | 13.6       | 79.8  | 11.2       | 81.2  | 8.1        | 83.0  |
|                   |      |     | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |     | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 58.8  | 39.0  | 40.2  | 48.1  | 56.7  | 66.9  | 75.2  | 79.5  | 79.1  | 70.3  | 59.4  | 48.5  | 40.9 | (6)  |
| (7)  |                                                                   | DBStd       | 15.93 | 6.02  | 7.24  | 6.47  | 7.12  | 5.05  | 5.22  | 4.92  | 5.81  | 5.96  | 6.78  | 6.86  | 6.14 | (7)  |
| (8)  |                                                                   | HDD50       | 1163  | 342   | 281   | 116   | 21    | 0     | 0     | 0     | 0     | 12    | 104   | 287   |      | (8)  |
| (9)  |                                                                   | HDD65       | 3777  | 805   | 694   | 525   | 260   | 36    | 1     | 0     | 1     | 20    | 194   | 494   | 747  | (9)  |
| (10) |                                                                   | CDD50       | 4363  | 2     | 7     | 56    | 222   | 525   | 756   | 915   | 903   | 609   | 302   | 60    | 6    | (10) |
| (11) |                                                                   | CDD65       | 1504  | 0     | 0     | 0     | 12    | 96    | 307   | 450   | 440   | 180   | 19    | 0     | 0    | (11) |
| (12) |                                                                   | CDH74       | 13700 | 0     | 0     | 1     | 56    | 537   | 2608  | 4631  | 4665  | 1155  | 47    | 0     | 0    | (12) |
| (13) |                                                                   | CDH80       | 5429  | 0     | 0     | 0     | 3     | 84    | 922   | 2003  | 2095  | 320   | 2     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.6   | 4.5   | 4.5   | 4.7   | 4.6   | 4.6   | 4.8   | 4.6   | 4.7   | 4.5   | 4.4   | 4.4   | (14) |      |
| (15) | Precipitation                                                     | PrecAvg     | 38.4  | 1.5   | 2.1   | 2.8   | 4.0   | 5.0   | 5.1   | 3.2   | 3.2   | 3.2   | 3.9   | 3.0   | 1.5  | (15) |
| (16) |                                                                   | PrecMax     | 58.5  | 5.1   | 4.2   | 5.5   | 10.0  | 12.0  | 11.7  | 9.4   | 11.8  | 10.0  | 8.6   | 6.6   | 3.5  | (16) |
| (17) |                                                                   | PrecMin     | 20.8  | 0.2   | 0.2   | 0.8   | 0.6   | 1.4   | 1.1   | 0.0   | 0.0   | 0.4   | 0.2   | 0.4   | 0.1  | (17) |
| (18) |                                                                   | PrecStd     | 7.6   | 0.9   | 1.0   | 1.1   | 2.2   | 2.6   | 3.0   | 2.6   | 3.1   | 2.4   | 2.0   | 1.7   | 1.0  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 58.8  | 62.3  | 69.4  | 79.6  | 86.0  | 94.4  | 97.7  | 98.3  | 90.5  | 78.5  | 67.9  | 61.0 | (19) |
| (20) |                                                                   |             | MCWB  | 45.0  | 47.8  | 52.2  | 61.2  | 64.6  | 69.8  | 71.6  | 69.6  | 71.1  | 67.5  | 55.5  | 48.3 | (20) |
| (21) |                                                                   | 2%          | DB    | 54.4  | 57.6  | 65.4  | 75.2  | 82.2  | 90.8  | 94.8  | 95.2  | 87.3  | 74.9  | 63.8  | 55.2 | (21) |
| (22) |                                                                   |             | MCWB  | 43.5  | 46.1  | 51.1  | 59.6  | 63.9  | 68.7  | 71.2  | 69.9  | 69.3  | 63.5  | 53.8  | 45.5 | (22) |
| (23) |                                                                   | 5%          | DB    | 51.0  | 53.8  | 61.9  | 71.9  | 79.2  | 87.9  | 92.1  | 92.6  | 83.7  | 71.8  | 60.7  | 51.8 | (23) |
| (24) |                                                                   |             | MCWB  | 41.4  | 44.3  | 49.5  | 57.9  | 63.3  | 68.3  | 70.7  | 70.0  | 67.6  | 62.4  | 52.1  | 43.9 | (24) |
| (25) |                                                                   | 10%         | DB    | 47.5  | 50.0  | 58.5  | 68.3  | 76.8  | 85.4  | 89.6  | 90.2  | 80.3  | 68.9  | 57.6  | 48.9 | (25) |
| (26) |                                                                   |             | MCWB  | 39.7  | 42.4  | 47.9  | 56.0  | 62.7  | 67.5  | 70.2  | 69.2  | 66.2  | 60.6  | 50.3  | 42.8 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 46.6  | 50.2  | 55.3  | 63.6  | 68.4  | 72.7  | 76.6  | 79.0  | 73.8  | 70.2  | 60.7  | 49.9 | (27) |
| (28) |                                                                   |             | MDCB  | 57.1  | 57.3  | 65.5  | 76.2  | 81.0  | 88.0  | 90.8  | 87.7  | 81.8  | 72.9  | 62.7  | 55.8 | (28) |
| (29) |                                                                   | 2%          | WB    | 44.1  | 47.3  | 52.6  | 61.0  | 66.3  | 70.9  | 74.3  | 74.3  | 71.5  | 66.3  | 55.7  | 46.8 | (29) |
| (30) |                                                                   |             | MDCB  | 52.6  | 56.3  | 62.2  | 73.3  | 77.8  | 85.7  | 88.8  | 87.8  | 83.2  | 72.8  | 60.9  | 53.0 | (30) |
| (31) |                                                                   | 5%          | WB    | 42.2  | 44.9  | 50.7  | 58.9  | 64.8  | 69.5  | 72.9  | 72.2  | 69.6  | 63.6  | 53.2  | 45.0 | (31) |
| (32) |                                                                   |             | MDCB  | 49.0  | 53.0  | 59.5  | 69.6  | 76.1  | 84.6  | 87.3  | 86.9  | 81.9  | 70.0  | 58.6  | 50.3 | (32) |
| (33) |                                                                   | 10%         | WB    | 40.5  | 42.7  | 49.1  | 56.9  | 63.3  | 68.3  | 71.5  | 70.7  | 67.7  | 61.2  | 51.2  | 43.3 | (33) |
| (34) |                                                                   |             | MDCB  | 46.0  | 48.5  | 57.1  | 66.1  | 74.4  | 83.3  | 85.7  | 85.7  | 78.4  | 67.7  | 56.8  | 48.1 | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 11.3  | 11.2  | 14.3  | 15.5  | 16.1  | 17.0  | 16.8  | 17.1  | 15.9  | 13.1  | 12.2  | 11.6 | (35) |
| (36) |                                                                   |             | MCDBR | 17.1  | 18.3  | 20.1  | 20.2  | 19.8  | 18.9  | 19.2  | 20.2  | 19.7  | 17.3  | 17.6  | 15.8 | (36) |
| (37) |                                                                   |             | MCWBR | 10.3  | 10.3  | 10.7  | 10.4  | 9.8   | 8.7   | 8.2   | 8.1   | 8.8   | 10.1  | 11.1  | 10.4 | (37) |
| (38) |                                                                   | 5% WB       | MCDBR | 15.3  | 16.8  | 17.7  | 18.4  | 17.0  | 17.7  | 17.1  | 18.0  | 17.9  | 15.6  | 15.3  | 13.7 | (38) |
| (39) |                                                                   |             | MCWBR | 9.8   | 10.1  | 9.8   | 10.2  | 9.9   | 9.1   | 8.3   | 9.2   | 9.6   | 10.8  | 10.5  | 9.2  | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.359 | 0.410 | 0.484 | 0.563 | 0.531 | 0.521 | 0.582 | 0.563 | 0.479 | 0.443 | 0.385 | 0.356 | (40) |      |
| (41) |                                                                   | taud        | 2.146 | 1.991 | 1.860 | 1.748 | 1.877 | 1.951 | 1.834 | 1.873 | 2.048 | 2.078 | 2.170 | 2.185 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 239   | 242   | 238   | 229   | 242   | 244   | 227   | 227   | 238   | 229   | 227   | 228   | (42) |      |
| (43) |                                                                   | Edh at Noon | 33    | 45    | 58    | 69    | 62    | 58    | 65    | 61    | 47    | 41    | 32    | 29    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 553   | 797   | 1066  | 1397  | 1749  | 2023  | 1971  | 1805  | 1398  | 916   | 648   | 502   | (44) |      |
| (45) |                                                                   | RadStd      | 60    | 106   | 100   | 180   | 119   | 138   | 150   | 180   | 123   | 101   | 69    | 61    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   | (46) |
|      |                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |      |
| (47) | Regional (14 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | (47) |

Nomenclature: See separate page